



**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani**  
**Pilani Campus**  
**AUGS/ AGSR Division**

**FIRST SEMESTER 2026-27**  
**COURSE HANDOUT**

**Date: 01.08.2026**

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

**Course No** : **EEE F435**  
**Course Title** : **Digital Image Processing**  
**Instructor-in-Charge** : **Pawan Kamalkishor Ajmera**

**1. Course Description:** This is the first course on digital image processing. It begins with an introduction to the fundamentals of digital images and discusses the various discrete transformations, which are extensively used in image processing. It then goes on to discuss the different image processing techniques such as image enhancement, automatic image classification, and recognition.

**2. Scope and Objective of the Course:** Digital Image Processing (DIP) involves the use of computer algorithms to perform operations on digital images to improve their quality, extract useful information, or prepare them for further analysis. The main objectives of DIP include enhancing image details for better visual interpretation, restoring corrupted images, compressing image data for efficient storage and transmission, and analyzing images for applications like medical diagnosis, satellite imaging, and object recognition. Its scope spans various fields such as healthcare, agriculture, surveillance, remote sensing, and artificial intelligence, making it a vital tool in both scientific research and everyday technology.

**3. Textbooks:**

1. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing, Pearson Education, 4<sup>th</sup> edition, 2021

**4. Reference Books:**

1. Milan Sonka, Vaclav Hlavac, Roger Boyle, Image Processing, Analysis, and Machine Vision, Thomson-Engineering, 4<sup>th</sup> edition, 2007.
2. Anil K Jain, Fundamentals of Digital Image Processing, Prentice Hall, 1<sup>st</sup> edition, 1988.
3. Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, Digital Image Processing using MATLAB, McGraw-Hill, 2<sup>nd</sup> edition, 2010.

**5. Course Plan:**

| Module No. | Lecture Session                                         | Reference                | Learning outcomes                                                                                                                           |
|------------|---------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|            | (1) Introduction to Digital Image Processing            |                          | Understand fundamental concepts of digital images, applications of image processing.                                                        |
| I          | (2-4) Digital Image Fundamentals                        | T1: Ch 2                 | Understand digitization of images, spatial and gray-level resolution, pixel relationships                                                   |
| II         | (5-8) Intensity Transformation and Histogram processing | T1: Ch 3<br>R3: 2.2, 2.3 | Apply point processing techniques; Enhance image contrast using histograms; Implement gray-level transformations                            |
| III        | (9-10) Spatial Filtering                                | T1: Ch 3<br>R3: 2.4, 2.5 | Implement spatial filters for smoothing and sharpening; Understand convolution operations; Evaluate the impact of spatial filters on images |



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|        |                                          |                                   |                                                                                                                                             |
|--------|------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| IV     | (11-15) Frequency Domain Processing      | T1: Ch 4<br>T1: Ch 7<br>R3: Ch 3  | Transform images between spatial and frequency domains; Analyze frequency components of an image; Apply frequency filters for enhancement   |
| V      | (16-18) Image Restoration                | T1: Ch 5<br>R3: Ch 4              | Identify types of image degradation; Restore images using filters; Apply geometric operations like translation and rotation                 |
| VI A   | (19-20) Image Compression-I              | T1: Ch 8<br>R3: Ch 8              | Understand principles of image compression;                                                                                                 |
| VI B   | (21-24) Image Compression-II             | T1: Ch 8<br>R3: Ch 8              | Understand different compression techniques;                                                                                                |
| VII    | (25-27) Morphological Image Processing   | T1: Ch 9<br>R3: Ch 9              | Apply morphological operations on binary images; Extract structural features from images; Design morphological pipelines for specific tasks |
| VIII A | (28-30) Image Segmentation: Edge-Based   | T1: Ch 10<br>R3: 10.1, 10.2, 10.3 | Detect and analyze edges in images; implementation of edge-based segmentation methods                                                       |
| VIII B | (31-33) Image Segmentation: Region-Based | T1: Ch 10<br>R3: 10.4, 10.5,      | Segment images using thresholding; Implement region-based segmentation methods                                                              |
| IX     | (34-37) Representation and Description   | T1: Ch 11<br>R3: Ch 11            | Represent shapes using descriptors; Extract quantitative features from image regions                                                        |
| X      | (38-40) Case Studies and Applications    | Notes                             | Appreciate real-world uses of DIP - Relate course content to industrial scenarios                                                           |

**6. Evaluation Scheme:**

| Component                 | Duration | Weightage (%) | Date & Time | Nature of component (Close Book/ Open Book) |
|---------------------------|----------|---------------|-------------|---------------------------------------------|
| Mid-Semester Test         | 90 Min.  | 30            |             | Close book                                  |
| Comprehensive Examination | 180 Min. | 40            |             | Close Book                                  |
| Assignment                |          | 15            |             | Open Book                                   |
| Project                   |          | 15            |             | Open Book                                   |

**7. Chamber Consultation Hour:** to be announced in the class.

**8. Notices:** All notices of this course will be displayed on Nalanda.

**9. Make-up Policy:** Make-up will be given *ONLY* in extremely genuine cases; IC should be informed.

**10. Note (if any):**

**Instructor-in-charge**  
**Course No. EEE F 435**